Chapter 79 — Reviews & Ratings

> **Part IX — Feature Architectures** · Complexity: **Standard** (≥25 interview questions)
>
> **Prerequisites:** Ch 72 (optimistic updates), Ch 31 (outbox), Ch 15 (idempotency), Ch 3 (SSOT), Ch 13 (paging), Ch 17 (uploads), Ch 73 (recommendations/ranking), Ch 23 (security). Pairs with Ch 71 (feed), Ch 32 (sync).
>
> **Reference surface in StreamCart:** product/content reviews and star ratings — writing a review (text + photos + stars) reliably, displaying a **server-ranked** list of reviews with aggregates (average, distribution), handling moderation and abuse, and keeping the user's own review consistent.

79.1 Problem Definition and Real-World Context

Reviews and ratings combine several book patterns onto a **user-generated-content (UGC)** feature: **submitting** a review reliably (text + optional photos + a star rating, offline-capable, idempotent); **displaying** a **server-ranked, paginated** list of reviews plus **server-computed aggregates** (average rating, star distribution, count); **moderation** (UGC is untrusted and may be abusive/spam); and keeping the **user's own review** consistent (edit/delete). The composition touches optimistic submit (Ch 72), outbox + idempotency (Ch 31/15), media upload (Ch 17), paging + ranking (Ch 13/73), SSOT (Ch 3), and UGC security/moderation (Ch 23).

Concretely, teams hit:

- **Unreliable submit.** A review submitted offline/flaky is lost or double-posted. It needs an **outbox** (Ch 31) + **idempotency key** (Ch 15) and optimistic display (Ch 72) with a "pending/posted/failed" status (like a comment, Ch 64).
- **Client-computed aggregates.** Computing the average rating / distribution on the client is wrong — aggregates span **all** reviews (which the client doesn't have) and must be **server-computed** (Ch 73). The client **displays** server aggregates.
- **Client-ranked reviews.** Review ordering ("most helpful", "recent", "critical") is a **server ranking** (helpfulness, recency, verified-purchase, Ch 73); the client renders the server's order + logs signals (helpful votes), it doesn't rank.
- **UGC is untrusted + needs moderation.** Reviews are user content → **render safely** (sanitize, Ch 52), **moderate** (spam/abuse/profanity — server-side), and handle **report/flag**. Photos need upload + moderation (Ch 17).
- **The user's own review consistency.** The author's review (edit/delete, pending state) must be consistent in the SSOT (Ch 3) and reflected in the list; "you already reviewed this" logic.
- **Media in reviews.** Review photos upload reliably (Ch 17) with progress; the review posts referencing uploaded media.
- **Verified purchase / eligibility.** Often only **purchasers** can review (server- enforced, Ch 23); "verified purchase" badges come from the server.

The architecture: **submit** a review optimistically via an **outbox + idempotency key** (Ch 72/31/15) with **media upload** (Ch 17) and a pending/posted/failed status; **display** a **server-ranked, paginated** review list (Ch 13/73) with **server-computed aggregates** (average/distribution/count); **moderate** UGC server-side and **render untrusted content safely** (Ch 52/23); keep the **user's own review** consistent in the SSOT (edit/delete, Ch 3); and enforce **eligibility** (verified purchase) server-side. This chapter composes the UGC-submit + ranked-list + aggregates + moderation stack.

Three practical use cases

1. **Write a review (StreamCart).** Stars + text + photos → optimistic "posting…" → outbox + idempotent submit (Ch 31/15) + photo upload (Ch 17) → "posted" (or "failed" + retry); eligibility (verified purchase) enforced server-side.
2. **Review list + aggregates.** A product's reviews, **server-ranked** ("most helpful"), **paginated** (Ch 13), with the **server-computed** average (4.3★), distribution, and count; "helpful" votes are optimistic signals (Ch 72/73).
3. **Moderation & reporting.** Submitted reviews pass **server moderation**; users can **report** abusive reviews; rendered content is **sanitized** (Ch 52).

Why this is architectural

Reviews compose **optimistic UGC submit** (Ch 72/31/15/17), **server-ranked paginated lists + server aggregates** (Ch 13/73), **SSOT** (Ch 3), and **UGC security/moderation** (Ch 23/52). The decisions — reliable idempotent submit, server-computed aggregates, server-ranked list, moderation + safe rendering, verified-purchase eligibility — recur for comments, Q&A, and any UGC feature.

79.2 Requirements Engineering

Functional requirements

ID	Requirement
F1	Submit a review (stars + text + photos) reliably: optimistic + outbox + idempotent (Ch 72/31/15).
F2	Upload review photos reliably with progress (Ch 17); review references uploaded media.
F3	Display a server-ranked, paginated review list + server-computed aggregates (Ch 13/73).
F4	Moderate UGC server-side; render untrusted content safely; report/flag (Ch 23/52).
F5	Keep the user's own review consistent (edit/delete, pending state) in the SSOT (Ch 3).
F6	Enforce eligibility (verified purchase) server-side; "helpful" votes as signals (Ch 23/73).

Non-functional requirements

Attribute	Policy
Reliability	Submit survives offline/death; idempotent; media uploaded.
Correctness	Server-computed aggregates; server-ranked list; no client math/ranking.
Safety	UGC moderated + sanitized; eligibility + abuse handled (Ch 23).
Consistency	User's own review consistent (SSOT); list reflects it (Ch 3).
Scale	Paged list; aggregates efficient (server).

Scope

- **In scope:** review submit (optimistic + outbox + idempotency + media upload, Ch 72/31/ 15/17), server-ranked paginated list (Ch 13/73), server-computed aggregates (average/ distribution/count), moderation (server) + safe rendering (Ch 52) + report/flag, user's own review consistency (edit/delete/pending, Ch 3), eligibility (verified purchase, Ch 23), "helpful" votes (optimistic signals, Ch 72/73).
- **Out of scope:** optimistic-update mechanics (Ch 72), outbox/idempotency (Ch 31/15), upload (Ch 17), paging/ranking internals (Ch 13/73), SSOT (Ch 3), moderation ML/backend.

Assumptions & constraints

- Room SSOT (Ch 3), outbox (Ch 31), upload (Ch 17), Paging 3 (Ch 13). The **server** ranks reviews, computes aggregates, moderates, and enforces eligibility (Ch 73/23). `minSdk 24`. UGC is untrusted; aggregates/ranking are server-owned.

79.3 System Design Fundamentals

- **Submit = reliable optimistic UGC create (Ch 72/31/15/17).** Writing a review is a **create** with media:
  - **Optimistic insert** into the SSOT with a **client-generated id** and status **Posting** (Ch 72/64) → the user sees their review immediately.
  - **Upload photos** reliably (Ch 17) with progress; the review references the **uploaded media ids**.
  - **Submit via the outbox** (Ch 31) with the **client id as the idempotency key** (Ch 15) so offline/death-safe, retried, and **not double-posted**.
  - On success → **Posted** (reconcile to the server id/timestamp); on genuine failure → **Failed** + retry (Ch 72). Same pattern as a comment (Ch 64).
- **Aggregates are server-computed (not client, Ch 73).** The **average rating**, **star distribution** (how many 5★/4★/…), and **count** are over **all** reviews — the client **doesn't have them all**, so it **cannot** compute aggregates correctly. The **server computes** aggregates; the client **displays** them (and updates optimistically with caution, then reconciles — a single new review barely moves a 10k-review average, so show the server value). Client-computed averages are wrong and drift.
- **Reviews are a server-ranked, paginated list (Ch 13/73).** Review **ordering** — "most helpful", "most recent", "critical", "verified" — is a **server ranking** (helpfulness votes, recency, verified-purchase, length, Ch 73). The client **renders the server's order**, **pages** it (Paging 3 + SSOT, Ch 13/71), and **logs signals** (helpful

votes) — it **does not rank**. This reuses the feed infra (Ch 71) and the recommendations division of labor (Ch 73).

- **"Helpful" votes = optimistic signals (Ch 72/73).** Voting a review helpful is an **optimistic toggle** (Ch 72) that also **feeds the server's ranking signal** (Ch 73) — like a like (Ch 72). The displayed helpful count is server-reconciled.
- **UGC is untrusted → moderate + render safely (Ch 23/52).**
  - **Moderation (server-side):** reviews pass **server moderation** (spam/abuse/profanity/ policy — often ML + human), which may hold a review as **pending moderation** before it appears publicly. The author sees their own pending review; others don't until approved.
  - **Safe rendering (Ch 52):** review text/links are **untrusted** — **sanitize** (no injection, safe links), handle media safely; never auto-execute content.
  - **Report/flag:** users can **report** a review; reports go to server-side moderation.
- **The user's own review is consistent (Ch 3).** The author's review lives in the **SSOT** (one row) with its status (pending/posted/failed/moderation); **edit/delete** are idempotent mutations (Ch 15) via the outbox (Ch 31); the list shows "your review" and "you already reviewed this" from the SSOT. Editing updates the row everywhere.
- **Eligibility: verified purchase (server-enforced, Ch 23).** Often only **purchasers** may review; the **server enforces** eligibility (checks the order history) — the client shows/hides the review CTA based on a server-provided flag, but the **server authorizes** the submit (a client can't bypass). "Verified purchase" badges come from the server.
- **Media moderation + upload (Ch 17).** Review photos **upload** reliably (Ch 17, resumable + progress) and are **moderated** server-side (inappropriate images) before appearing.

Core insight: **reviews are a user-generated-content feature composing reliable optimistic submit (optimistic SSOT insert + photo upload + outbox with an idempotency key so it's offline/death-safe and not double-posted, Ch 72/31/15/17) with a server-ranked, paginated review list and server-computed aggregates (average/distribution/count — the client displays, never computes or ranks, Ch 13/73), server-side moderation and safe rendering of untrusted content (Ch 23/52) with report/flag, the user's own review kept consistent in the SSOT (edit/delete, pending/posted/failed status, Ch 3), server-enforced eligibility (verified purchase, Ch 23), and "helpful" votes as optimistic ranking signals (Ch 72/73) — the server ranks/aggregates/moderates/authorizes, the client submits/displays/signals.**

79.4 High-Level Architecture



Components & ownership

- **Submit:** eligibility (server) → optimistic SSOT insert → photo upload → outbox (idempotent) → server moderates/persists/ranks/aggregates.
- **Display:** server-ranked paginated list + server-computed aggregates; untrusted content rendered safely.
- **Server owns** ranking, aggregates, moderation, eligibility; **client submits/displays/ signals**; user's own review consistent in the SSOT.

Direction: write → eligibility → optimistic + upload + outbox(idempotent) → server (moderate/rank/aggregate) → status/SSOT; display server-ranked list + server aggregates safely.

79.5 Detailed Component Design



Component	Responsibility
`Review`	SSOT row: client/server id + rating/text/media + status + helpful + verified (Ch 3)
`ReviewRepository`	submit (optimistic+upload+outbox), ranked list, aggregates, vote, report
`Aggregates`	server-computed average/distribution/count (client displays)

79.6 End-to-End Data Flow

Submit a review (optimistic + upload + idempotent)



Display: server-ranked list + server aggregates



79.7 Complete Kotlin Implementation

Package: `com.streamcart.reviews`.

79.7.1 Review SSOT (client/server id + status, Ch 3/15)

```
@Entity(tableName = "reviews")
data class Review(
    @PrimaryKey val clientId: String,        // idempotency key + optimistic match (Ch 15/64)
    val serverId: String?, val productId: String, val authorId: String,
    val rating: Int, val text: String, val mediaIds: List<String>,
    val status: Status, val helpfulCount: Int, val verifiedPurchase: Boolean, val createdAt:
Long,
) { enum class Status { Posting, Posted, PendingModeration, Failed } }
```

79.7.2 Submit: optimistic + upload + outbox (idempotent, Ch 72/17/31/15)

```
@Singleton
class ReviewRepository @Inject constructor(
    private val dao: ReviewDao, private val uploader: MediaUploader,   // Ch 17
    private val outbox: Outbox, private val session: SessionManager,   // Ch 31/19
) {
    suspend fun submit(productId: String, rating: Int, text: String, photos: List<Uri>) {
        val clientId = UUID.randomUUID().toString()
        dao.upsert(Review(clientId, serverId = null, productId, session.userId, rating, text,
            mediaIds = emptyList(), status = Status.Posting, helpfulCount = 0,
            verifiedPurchase = false, createdAt = System.currentTimeMillis()))   // optimistic
(Ch 72)
        val mediaIds = photos.map { uploader.upload(it) }                         // reliable
upload + progress (Ch 17)
        dao.setMedia(clientId, mediaIds)
        outbox.enqueue(OutboxOp.SubmitReview(clientId, productId, rating, text, mediaIds))   //
idempotent (Ch 15/31)
    }

    suspend fun sync(op: OutboxOp.SubmitReview): OutboxResult = runCatching {
        val result = api.submitReview(op.clientId, op.productId, op.rating, op.text,
op.mediaIds)  // authed + eligible (Ch 23)
        dao.confirm(op.clientId, serverId = result.id, status = result.statusForClient())   //
Posted / PendingModeration
        OutboxResult.Done
    }.getOrElse { e ->
        if (e.isTransient()) OutboxResult.Retry                          // offline/5xx -> retry
(Ch 2/29)
        else { dao.setStatus(op.clientId, Status.Failed); OutboxResult.Failed(e) }   // genuine
-> Failed + retry UX (Ch 72)
    }
}
```

79.7.3 Server-ranked, paginated list (Ch 13/73) + safe rendering (Ch 52)

```
fun ReviewRepository.reviews(productId: String, sort: ReviewSort): Flow<PagingData<Review>> =
    Pager(PagingConfig(pageSize = 20), remoteMediator = ReviewRemoteMediator(productId, sort,
api, db)) {
        db.reviews().pagingSource(productId)                // server-RANKED order (Ch 73);
client renders (Ch 13/71)
    }.flow
// RemoteMediator fetches server-ranked pages (sort = Helpful/Recent/Critical) -> Room. Client
does NOT rank.

@Composable
fun ReviewText(text: String) {
    Text(sanitizeUgc(text))   // UNTRUSTED content -> sanitize (Ch 52); safe links; no injection
}
```

79.7.4 Aggregates: server-computed, client displays (Ch 73)

```
data class Aggregates(val average: Double, val distribution: Map<Int, Int>, val count: Int)  //
SERVER-computed

fun ReviewRepository.aggregates(productId: String): Flow<Aggregates> =
    db.aggregates().observe(productId)                       // client DISPLAYS server
aggregates (Ch 3)
// Refreshed from the server (api.reviewAggregates(productId)); NEVER computed client-side
(needs all reviews).

@Composable
fun RatingSummary(agg: Aggregates) {
    Text("${agg.average} ★ (${agg.count})")                  // display server values
    agg.distribution.forEach { (stars, n) -> StarBar(stars, n) }
}
```

79.7.5 Helpful vote (optimistic signal) + report (Ch 72/73)

```
suspend fun ReviewRepository.voteHelpful(reviewId: String) {
    dao.incrementHelpfulOptimistic(reviewId)                 // optimistic (Ch 72) -> instant
    outbox.enqueue(OutboxOp.HelpfulVote(reviewId, idempotencyKey = "$reviewId:helpful"))  //
idempotent (Ch 15); server ranking signal (Ch 73)
}
suspend fun ReviewRepository.report(reviewId: String, reason: String) =
    api.reportReview(reviewId, reason)                       // -> server moderation
```

79.7.6 Eligibility (server-enforced, Ch 23)

```
// Client shows/hides the CTA from a server flag; the SERVER authorizes the submit (verified
purchase).
fun ReviewRepository.canReview(productId: String): Flow<Boolean> =
db.eligibility().observe(productId)
// Server checks order history -> eligible; a client can't bypass (submit is authorized server-
side, Ch 23).
```

79.7.6 Module structure

```
:feature:reviews  Review/Dao, ReviewRepository (submit/list/aggregates/vote/report),
ReviewRemoteMediator
:core:upload  MediaUploader (Ch 17); :core:outbox (Ch 31); :core:data SSOT (Ch 3)
# Server: ranking + aggregates + moderation + eligibility (Ch 73/23). Optimistic -> Ch 72;
paging -> Ch 13; sanitize -> Ch 52.
```

79.8 Concurrency and Race Conditions

Scenario	Risk	Handling
Submit offline/double-tap	lost/double review	outbox + idempotency key (Ch 31/15)
Client computes average	wrong aggregate	server-computed; client displays (Ch 73)
Client ranks reviews	wrong order	server-ranked; client renders (Ch 73)
Edit own review	inconsistent	SSOT one row + idempotent edit (Ch 3/15)
Helpful double-vote	inflated	idempotent vote (Ch 15)
Untrusted content	injection	sanitize render (Ch 52)
Pending moderation shown to others	policy	server gates visibility; author sees own

- **Server owns aggregates and ranking** (Ch 73): the client **cannot** correctly compute an average over all reviews or rank by helpfulness — it **displays** server values and **renders** the server order, logging helpful votes as signals. Client-side math/ranking is wrong and drifts.
- **Reliable optimistic submit** (Ch 72/31/15/17): optimistic insert + photo upload + outbox with an idempotency key make submit offline/death-safe and dedup double-taps (same as comments, Ch 64).
- **UGC untrusted** (Ch 52): sanitize all rendered review text/links; moderation gates visibility server-side (author sees their own pending review; others don't).

79.9 Failure Scenarios and Recovery

Failure	Detection	Response	Outcome
Review lost offline/on death	missing	outbox persisted + retry (Ch 31/29)	not lost
Double-post on retry	duplicate	idempotency key (Ch 15)	single
Wrong average	mismatch	server-computed aggregates (Ch 73)	correct
Wrong review order	UX	server-ranked (Ch 73)	correct
Abusive/spam review	report/moderation	server moderation + report/flag (Ch 23)	handled
Injection via content	security	sanitize render (Ch 52)	safe
Ineligible submit	server reject	server-enforced eligibility (Ch 23)	blocked
Photo upload fails	incomplete review	resumable upload + retry (Ch 17)	completes


79.9 Failure Scenarios and Recovery

> **Reviews = reliable optimistic UGC submit + server-owned aggregates/ranking/moderation.** The client submits (optimistic + upload + outbox + idempotency), displays (server aggregates, server-ranked list, safely rendered), and signals (helpful votes) — the server ranks, aggregates, moderates, and authorizes. Client-computed averages/rankings and unreliable submits are the recurring bugs.

79.10 Scaling the Implementation

- **Level 1.** Submit (optimistic + outbox + idempotency, Ch 72/31/15); display server- ranked paginated list (Ch 13/73) + server aggregates.
- **Level 2.** Photo upload (Ch 17); helpful votes (optimistic signals, Ch 72/73); sanitize rendering (Ch 52); user's own review consistency (edit/delete, Ch 3).
- **Level 3 — production.** Server moderation + pending state; report/flag; verified- purchase eligibility (Ch 23); media moderation.
- **Level 4 — large.** Rich sorting/filtering (server-ranked), review search, author profiles, "was this helpful" ML signals (Ch 73), translation, incentivized-review controls.
- **Level 5 — enterprise.** Moderation pipeline governance, abuse/fraud detection (fake reviews), review analytics, aggregate-consistency SLOs, regulatory (disclosure).

Avoid overengineering: reliable submit + server aggregates/ranking + moderation + safe rendering is the core. Don't compute aggregates or rank on the client; don't fire-and-forget submits.

79.11 Alternative Approaches and Trade-offs

Dimension	Option A	Option B	When
Submit	Optimistic + outbox + idempotent (Ch 72/31/15)	POST-and-wait	Optimistic+outbox (offline/instant/no-dup)
Aggregates	Server-computed (Ch 73)	client-computed	Server (needs all reviews; client drifts)
Ranking	Server-ranked (Ch 73)	client-sorted	Server (helpfulness/verified signals)
Content	Sanitize + server-moderate (Ch 52/23)	trust UGC	Moderate+sanitize (untrusted)
Eligibility	Server-enforced (verified purchase)	client-only check	Server (bypassable client)
Own review	SSOT one row (Ch 3)	per-screen	SSOT (consistent edit/status)
Helpful vote	Optimistic signal (Ch 72/73)	wait-for-server	Optimistic (instant) + server signal

Server-computed aggregates vs client-computed: the average rating and distribution span **all** reviews, which the client doesn't have — computing them client-side is **wrong** and drifts as new reviews arrive. The **server computes** aggregates (over the full set); the client **displays** them (a single optimistic review barely moves a large average, so show the server value). Never compute review aggregates on the client.

Server-ranked vs client-sorted reviews: review ordering ("most helpful", "verified", "recent") depends on **global signals** (helpfulness votes across users, verified-purchase, recency) the client lacks — like recommendations (Ch 73), the **server ranks** and the client **renders + logs signals** (helpful votes). Client sorting by local data gives a poor, inconsistent order.

Optimistic+outbox vs POST-and-wait submit: a POST-and-wait submit shows a spinner, loses the review offline, and risks double-post on retry. **Optimistic insert + outbox + idempotency** (Ch 72/31/15) makes submit feel instant, work offline, survive death, and never double-post — the same reliable-create pattern as comments (Ch 64).

79.12 Security & Privacy Considerations

- **UGC is untrusted (Ch 52/23): sanitize** review text/links on render (no injection, safe links); validate/limit media (Ch 17); never auto-execute content.
- **Server-side moderation (Ch 23):** reviews pass server moderation (spam/abuse/profanity/ policy) — the client can't be the moderation authority; hold **pending** reviews from public view until approved (author sees their own).
- **Eligibility + authorization (Ch 23):** the **server enforces** who can review (verified purchase) and authorizes submit/edit/delete — a client flag only shows/hides the CTA; the server is the gate.
- **Report/flag → moderation:** provide reporting; route to server-side moderation; handle abuse/retaliation.
- **Author identity/privacy (Ch 28):** show the author per policy (name/handle/anonymous); don't leak PII; the user's own review is editable/deletable.
- **Fake-review/abuse detection (server):** velocity, device, and pattern checks are server-side; the client can't prevent fake reviews.
- **Per-account (Ch 42):** the user's own review state is per-account.
- Client vs backend: the **server** moderates, ranks, aggregates, enforces eligibility, and detects abuse; the **client** submits reliably, renders untrusted content safely, displays server aggregates/order, and logs signals.

79.13 Performance and Resource Optimization

- **Page the review list** (Ch 13/71) — never load all; server-ranked pages.
- **Display server aggregates** (cheap) — no client computation over large sets.
- **Reliable media upload** (Ch 17) — resumable, progress, right-sized images (Ch 56/57).
- **Optimistic submit/vote** — instant UX; reconcile in background.
- **Sanitize efficiently** — cache sanitized content where safe.
- **Right-sized review photos** (Ch 56/57) in the list.
- **Profiling:** submit success/latency, upload success, list render/jank (Ch 71), aggregate freshness, moderation hold rate.

79.14 Testing Strategy

Submit reliability, server-owned aggregates/ranking, and safe rendering are testable.

```
class ReviewSubmitTest {
    @Test fun `submit is optimistic, uploads media, enqueues idempotent op`() = runTest {
        repo.submit("p1", 5, "great", listOf(uri))
        assertEquals(Status.Posting, dao.last().status)            // optimistic (Ch 72)
        assertTrue(uploader.uploaded.isNotEmpty())                  // media (Ch 17)
        assertEquals(dao.last().clientId, outbox.last().idempotencyKey)  // idempotent (Ch 15)
    }
    @Test fun `offline submit retries, no double-post`() = runTest { /* outbox retry; dedup (Ch
31/15) */ }
    @Test fun `genuine failure -> Failed with retry`() = runTest { /* 4xx -> Failed (Ch 72) */ }
}

class ReviewDisplayTest {
    @Test fun `aggregates come from server, not computed`() = runTest {
        api.aggregates = Aggregates(4.3, mapOf(5 to 100), 123)
        assertEquals(4.3, repo.aggregates("p1").first().average, 0.0)   // displayed, not
computed (Ch 73)
    }
    @Test fun `list renders server order`() = runTest { /* server-ranked preserved (Ch 73) */ }
    @Test fun `review text is sanitized`() { assertFalse(sanitizeUgc("<script>").contains("
<script>")) }  // Ch 52
}
```

What to test: submit optimistic + upload + idempotent outbox + offline retry (no double-post)

- genuine-failure Failed; aggregates **displayed from server** (not computed); list renders **server order**; untrusted content **sanitized** (Ch 52); helpful vote optimistic + idempotent; eligibility gate (server). Fake API/uploader/outbox; Room in-memory.

79.15 Production Observability

- **Metrics:** submit success/latency + offline-queued; double-post rate (→ 0); media upload success; list render/jank (Ch 71); aggregate freshness; moderation hold/approval rate; report rate; helpful-vote reconcile; eligibility gate accuracy.
- **Logs:** `review{action, status, productId}` — no PII/content (Ch 28).
- **Debugging:** lost/double submit → outbox/idempotency (Ch 31/15); wrong average → client computing (server-compute, Ch 73); wrong order → client ranking; injection → not sanitizing (Ch 52); abusive content → moderation; ineligible submit allowed → server not enforcing (Ch 23).
- **Alerting:** double-post spike; moderation-queue backlog; abuse/report spike.
- **Dashboards:** submit funnel + success; moderation rates; aggregate freshness.

79.16 System Design Interview Preparation

> ≥25 questions.

Fundamentals

1. *Review submit?* Optimistic SSOT insert + photo upload + outbox with idempotency key (Ch 72/17/31/15).
2. *Offline submit?* Outbox persists + retries; no double-post (Ch 31/15).
3. *Status model?* Posting → Posted/PendingModeration/Failed.
4. *Who computes aggregates?* The server (average/distribution/count) — client displays (Ch 73).
5. *Why not client aggregates?* Needs all reviews; client drifts.

Display/ranking

6. *Review list?* Server-ranked, paginated (Ch 13/73); client renders + logs signals.
7. *Who ranks?* The server (helpfulness/verified/recency).
8. *Helpful vote?* Optimistic signal (Ch 72) feeding server ranking (Ch 73).
9. *Sort options?* Server-ranked variants (helpful/recent/critical).

UGC safety

10. *Content trust?* Untrusted — sanitize render (Ch 52).
11. *Moderation?* Server-side; pending reviews gated from public view.
12. *Report/flag?* To server moderation.
13. *Author sees pending?* Yes (own); others don't until approved.

Eligibility/consistency

14. *Who can review?* Server-enforced (verified purchase, Ch 23).
15. *Client flag?* Shows/hides CTA; server authorizes submit.
16. *Own review consistency?* SSOT one row + idempotent edit/delete (Ch 3/15).

Media

17. *Review photos?* Reliable upload (Ch 17); server media moderation.
18. *Verified-purchase badge?* Server-provided.

Security

19. *Fake reviews?* Server-side abuse detection.
20. *Injection?* Sanitize (Ch 52).
21. *Author privacy?* Per policy (Ch 28).

Testing/Production

22. *Test submit?* Optimistic + upload + idempotent + offline retry.
23. *Test aggregates?* Displayed from server (not computed).
24. *Lost/double submit?* Outbox/idempotency.
25. *Wrong average?* Client computing.
26. *Abuse?* Moderation + report.

Design-change challenges

1. **Submit a review with photos offline.** Optimistic + upload + outbox + idempotency (Ch 72/17/31/15).
2. **Show average rating + distribution.** Server-computed aggregates; client displays (Ch 73).
3. **Order reviews by "most helpful".** Server-ranked + helpful-vote signals (Ch 73).
4. **Handle abusive reviews.** Server moderation + report/flag + sanitize (Ch 23/52).
5. **Only purchasers can review.** Server-enforced eligibility (Ch 23).

Mock walkthrough (abridged)

Clarify (text/photos/stars, who can review, moderation?) → submit: optimistic + upload + outbox + idempotency (Ch 72/17/31/15) → display: server-ranked paginated list + server aggregates (Ch 13/73) → moderation + safe rendering + report (Ch 23/52) → own-review consistency (SSOT, Ch 3) → eligibility (server, Ch 23) → helpful votes (optimistic signals, Ch 72/73) → observability (submit funnel, moderation).

79.17 Practical Exercises

1. **Beginner.** Submit a star+text review optimistically via an outbox with an idempotency key; show Posting → Posted. *Eval:* offline-safe; no double-post.
2. **Intermediate.** Add photo upload (Ch 17) and display a server-ranked, paginated review list with server-computed aggregates. *Eval:* media uploaded; server order + aggregates.
3. **Advanced.** Add helpful votes (optimistic signals), report/flag, sanitized rendering, and pending-moderation state. *Eval:* votes instant + server signal; content safe; pending gated.
4. **Production debugging.** Fix a wrong average (client-computed) and a double-posted review (no idempotency). *Eval:* server aggregates; single review.
5. **System design challenge.** Enforce verified-purchase eligibility server-side and keep the user's own review consistent (edit/delete). *Eval:* ineligible blocked; own review consistent.

(Solutions withheld unless requested.)

79.18 Chapter Summary & Reference Sheet

Key decisions: treat reviews as a **user-generated-content** feature that **submits reliably** (optimistic SSOT insert + reliable **photo upload** + **outbox with a client-id idempotency key** so it's **offline/death-safe** and **never double-posted**, with Posting → Posted/PendingModeration/Failed status, Ch 72/31/15/17/64), **displays** a **server-ranked, paginated** review list (Ch 13/73) with **server-computed aggregates** (average/distribution/count — the client **displays, never computes or ranks**, Ch 73), **moderates** UGC server-side and **renders untrusted content safely** (sanitize, Ch 52) with **report/flag**, keeps the **user's own review consistent** in the SSOT (idempotent edit/delete, Ch 3), enforces **eligibility (verified purchase) server-side** (Ch 23), and treats **"helpful" votes as optimistic ranking signals** (Ch 72/73). The division of labor: the **server ranks/aggregates/moderates/authorizes/detects-abuse**; the **client submits, displays, renders safely, and signals**.

Concepts: reliable optimistic UGC submit (optimistic + upload + outbox + idempotency), server-ranked paginated list + server-computed aggregates, server moderation + safe rendering + report, user's own review consistency (SSOT), server-enforced eligibility, helpful votes as signals.

Common mistakes: fire-and-forget submit (lost offline, double-post); client-computed aggregates (wrong/drift); client-ranked reviews (wrong order); trusting/not sanitizing UGC (injection); no moderation/pending gating; client-only eligibility (bypassable); own review inconsistent (not SSOT); unreliable photo upload.

Checklist:

- [ ] submit optimistic + photo upload (Ch 17) + outbox + idempotency key (Ch 72/31/15)
- [ ] status Posting/Posted/PendingModeration/Failed
- [ ] server-ranked paginated list (Ch 13/73)
- [ ] server-computed aggregates (display only)
- [ ] sanitize untrusted content + server moderation + report (Ch 52/23)
- [ ] user's own review consistent in SSOT (edit/delete, Ch 3/15)
- [ ] server-enforced eligibility (verified purchase, Ch 23)
- [ ] helpful votes optimistic + server signal (Ch 72/73)
- [ ] per-account (Ch 42)
- [ ] submit-funnel + moderation telemetry.

Key APIs: Review SSOT (Room `@Entity` / `@Dao`, client/server id, Ch 3), Outbox + idempotency key (Ch 31/15), MediaUploader (Ch 17), Paging 3 + RemoteMediator (server-ranked, Ch 13/73), server aggregates endpoint (display), optimistic vote (Ch 72), `sanitizeUgc` (Ch 52), eligibility flag (server, Ch 23).

References:

- Offline-first mutations: https://developer.android.com/topic/architecture/data-layer/offline-first
- (composes) Ch 72 Optimistic Updates, Ch 31 Outbox, Ch 15 Idempotency, Ch 17 Uploads, Ch 13 Paging, Ch 73 Recommendations/Ranking, Ch 3 SSOT, Ch 52 Deep Links/Sanitize, Ch 23 Security, Ch 64 Chat

Chapter review (Phase 4)

- **Completeness:** all 18 sections; architecture/class/sequence/flow/failure diagrams; review SSOT + optimistic submit (upload+outbox+idempotency) + server-ranked list + server aggregates + helpful vote + eligibility code; concurrency (server-owned aggregates/ranking, reliable submit, UGC untrusted); failure (reliable-submit + server-owned principle); 5 levels; server-aggregates, server-ranking, optimistic-submit trade-offs; security (UGC untrusted, moderation, eligibility, abuse); perf (page/display-aggregates/upload); submit + aggregates + sanitize tests; observability (submit funnel, moderation); 26 Q&A + 5 challenges. ✅
- **Expand later:** optimistic (Ch 72), uploads (Ch 17), ranking (Ch 73), paging (Ch 13), moderation.
- **Interview coverage:** exceeds ≥25 bar.